

Watch-DevKit-HW

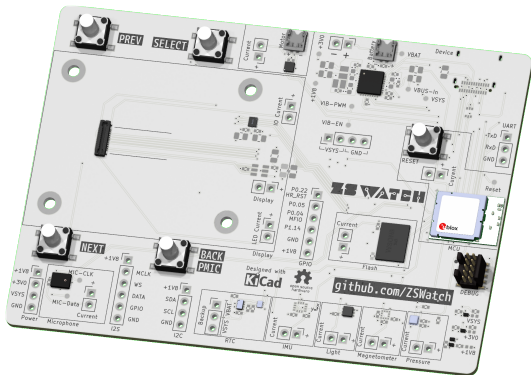
Variant: PRELIMINARY

2026-03-29

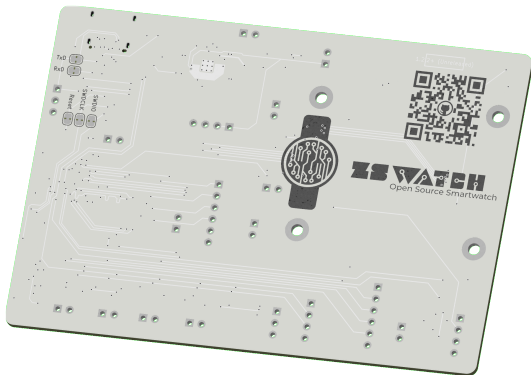
Rev 1.2.2+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 29-Mar-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

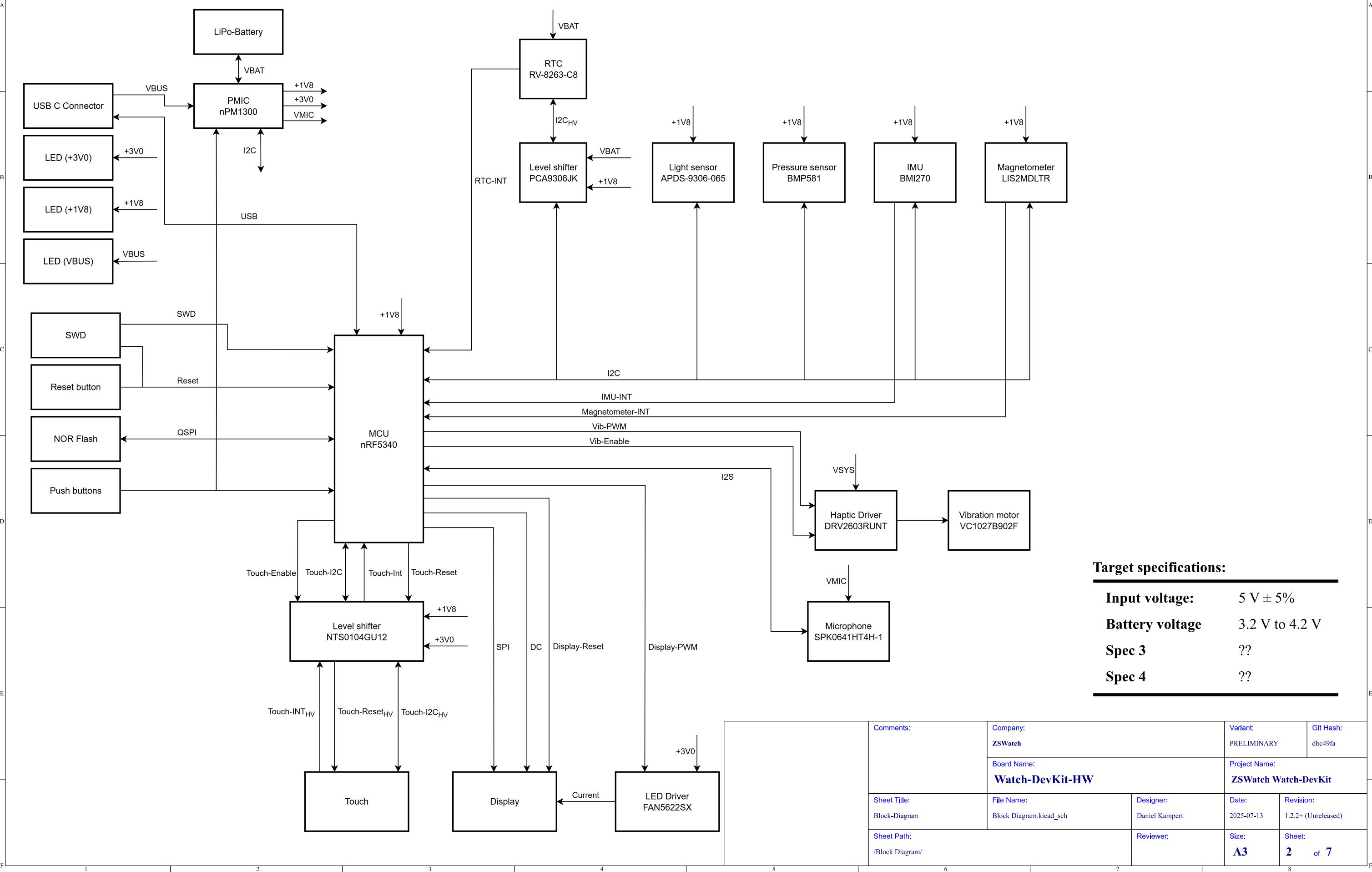
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

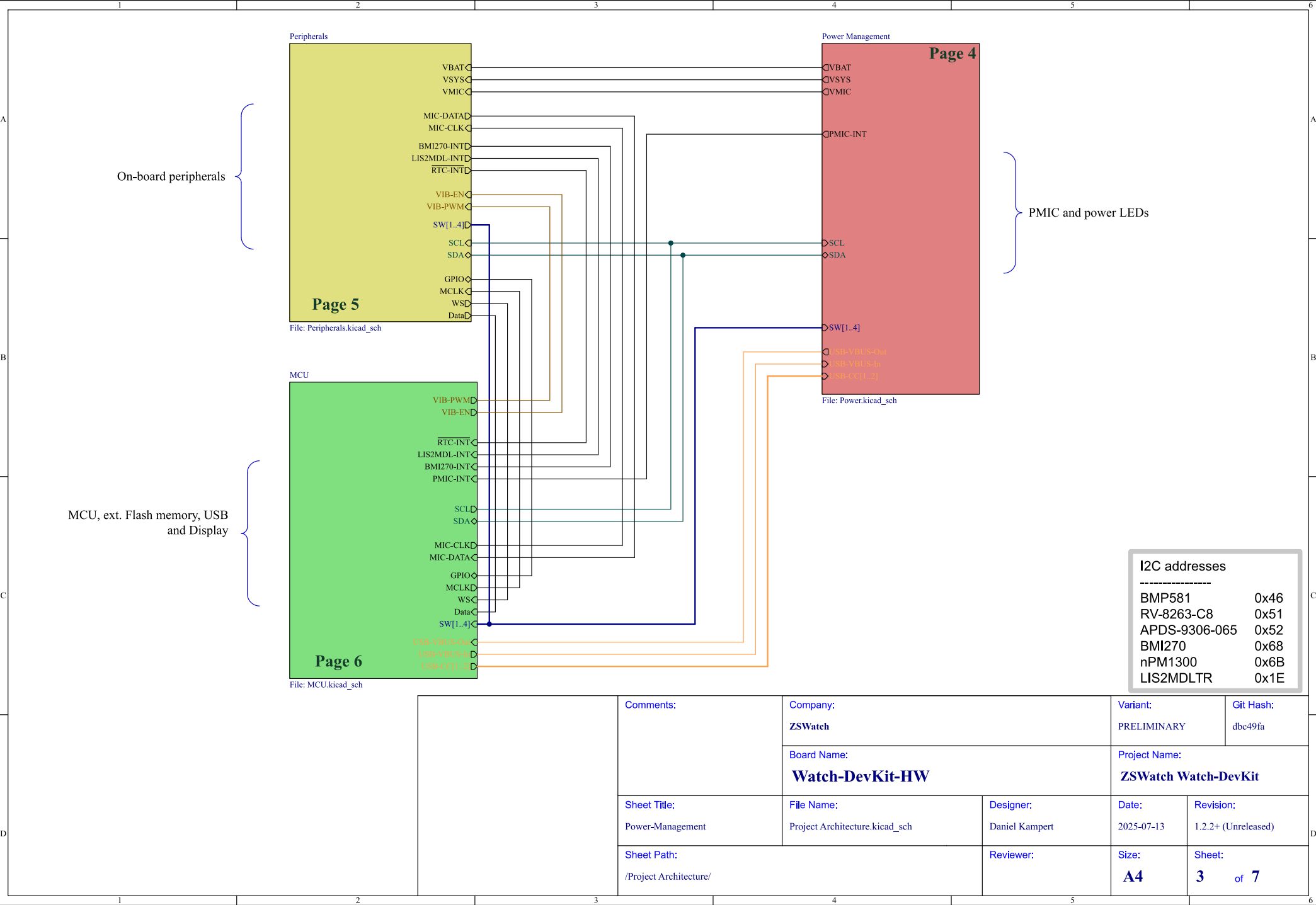
	Comments:	Company:	Variant:	Git Hash:
		ZSWatch	PRELIMINARY	dbc49fa
	Board Name:		Project Name:	
	Watch-DevKit-HW		ZSWatch Watch-DevKit	
Sheet Title:	File Name:	Designer:	Date:	Revision:
ZSWatch-Watch-DevKit	ZSWatch-Watch-DevKit.kicad_sch	Daniel Kampert	2025-07-13	1.2.2+ (Unreleased)
Sheet Path:	Reviewer:	Size:	Sheet:	
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[2] Block-Diagram



Target specifications:				
Input voltage:	5 V ± 5%			
Battery voltage	3.2 V to 4.2 V			
Spec 3	??			
Spec 4	??			

	Comments:	Company: ZSWatch		Variant: PRELIMINARY		Git Hash: dbc49fa		
		Board Name: Watch-DevKit-HW			Project Name: ZSWatch Watch-DevKit			
	Sheet Title: Block-Diagram	File Name: Block Diagram.kicad_sch		Designer: Daniel Kampert		Date: 2025-07-13	Revision: 1.2.2+ (Unreleased)	
	Sheet Path: /Block Diagram/			Reviewer:		Size: A3	Sheet: 2 of 7	



Comments:

Company:

ZSWatch

Variant:

PRELIMINARY

Git Hash:

dbc49fa

Board Name:

Watch-DevKit-HW

Project Name:

ZSWatch Watch-DevKit

Sheet Title:

Power-Management

File Name:

Project Architecture.kicad_sch

Designer:

Daniel Kampert

Date:

2025-07-13

Revision:

1.2.2+ (Unreleased)

Sheet Path:

/Project Architecture/

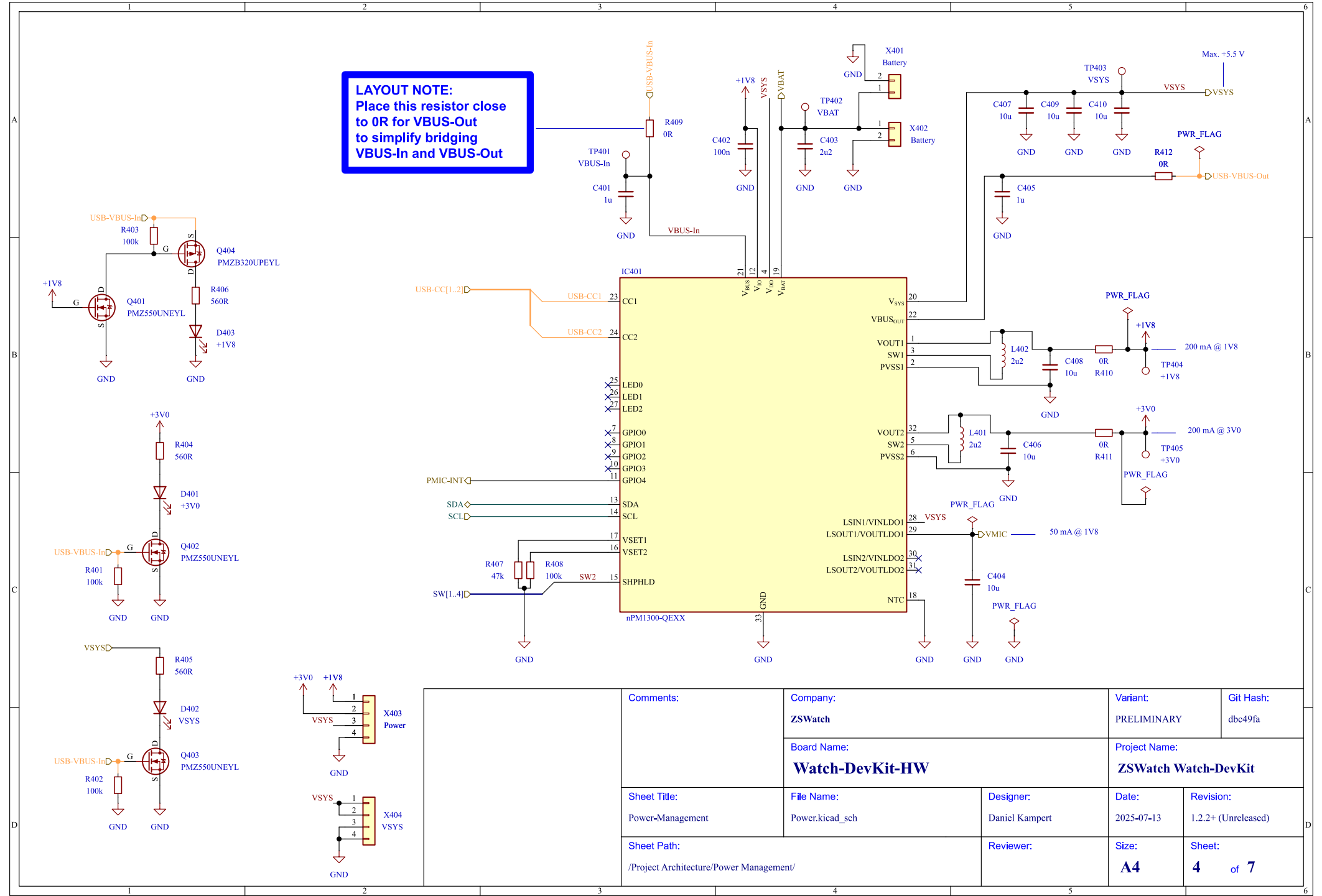
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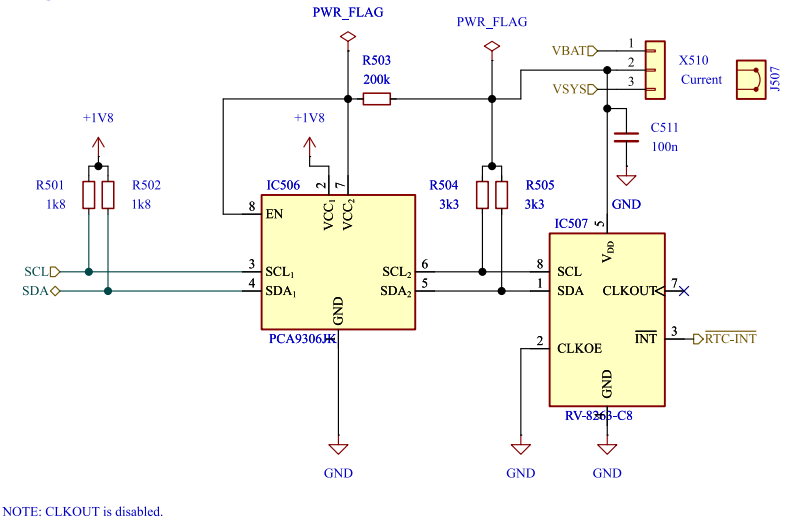
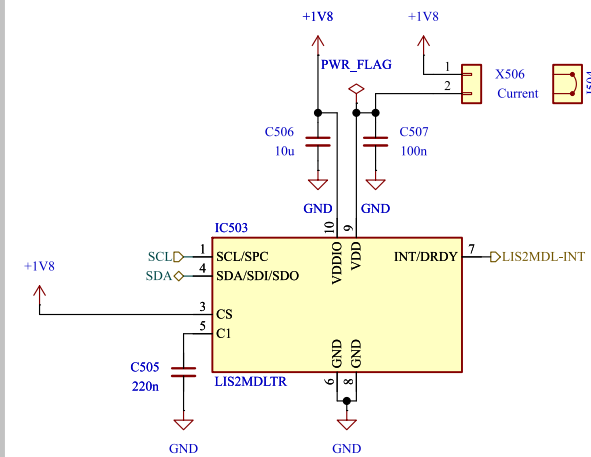
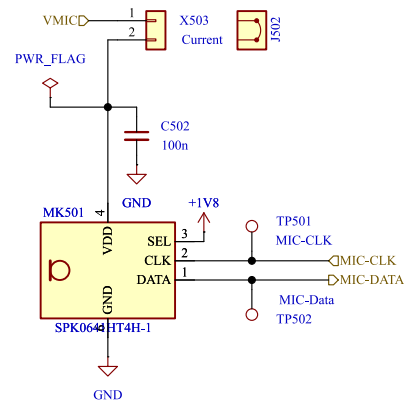
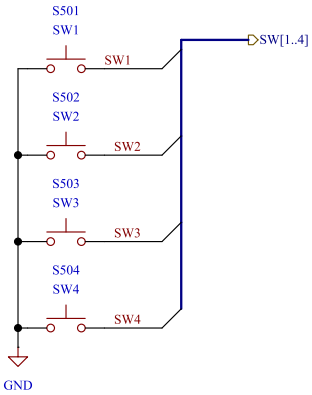
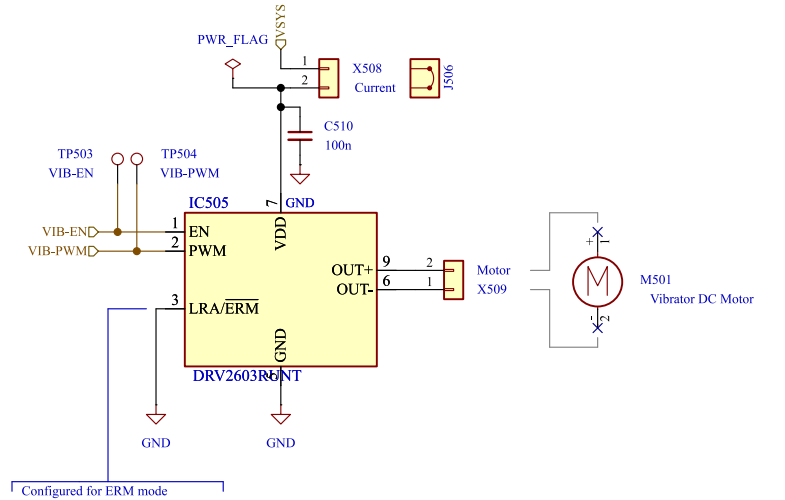
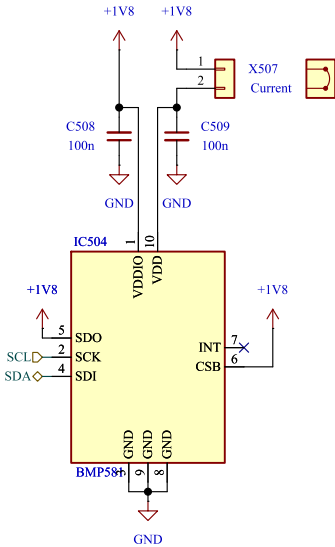
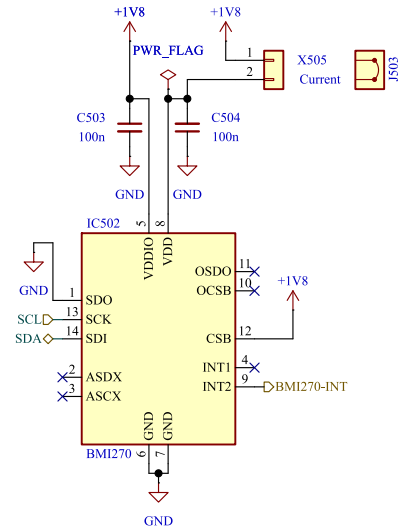
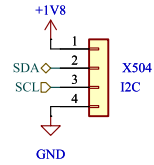
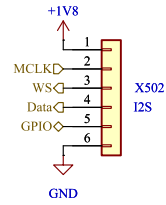
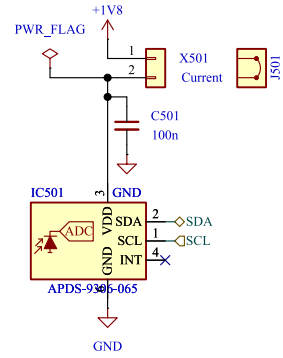
Size:

A4

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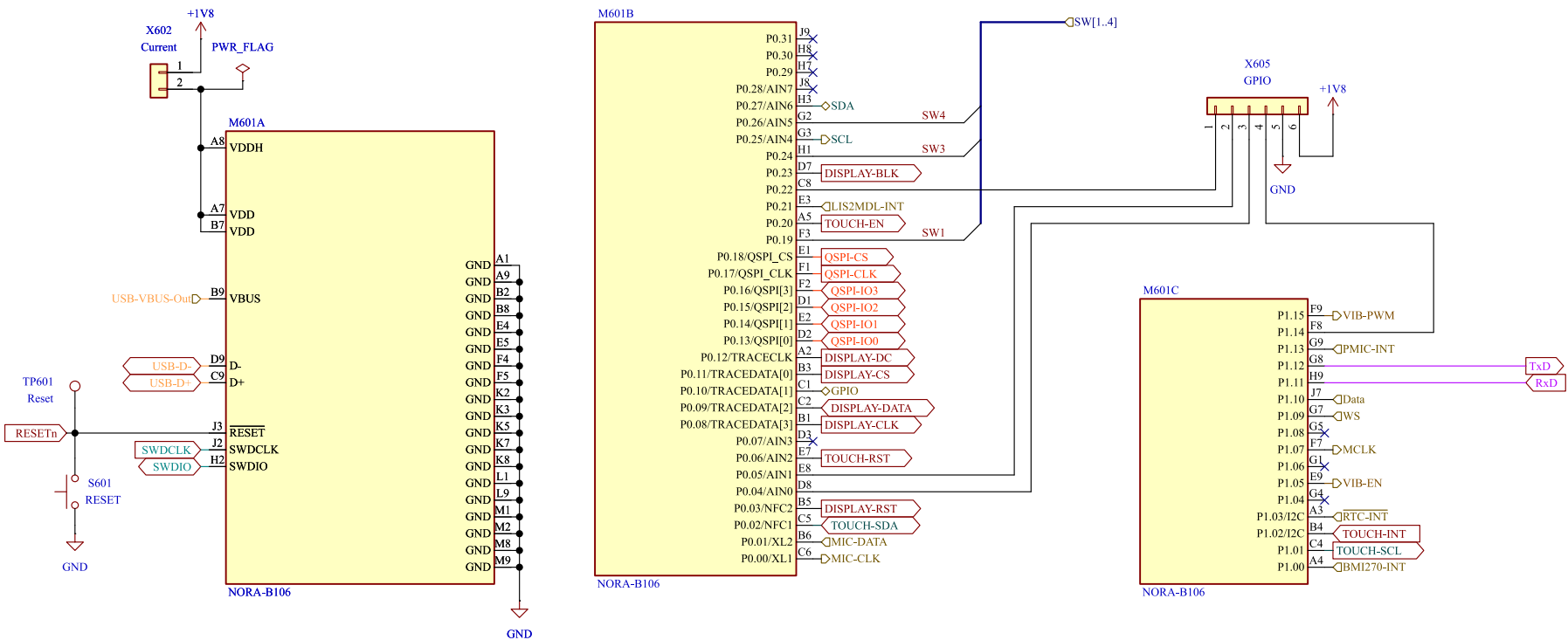
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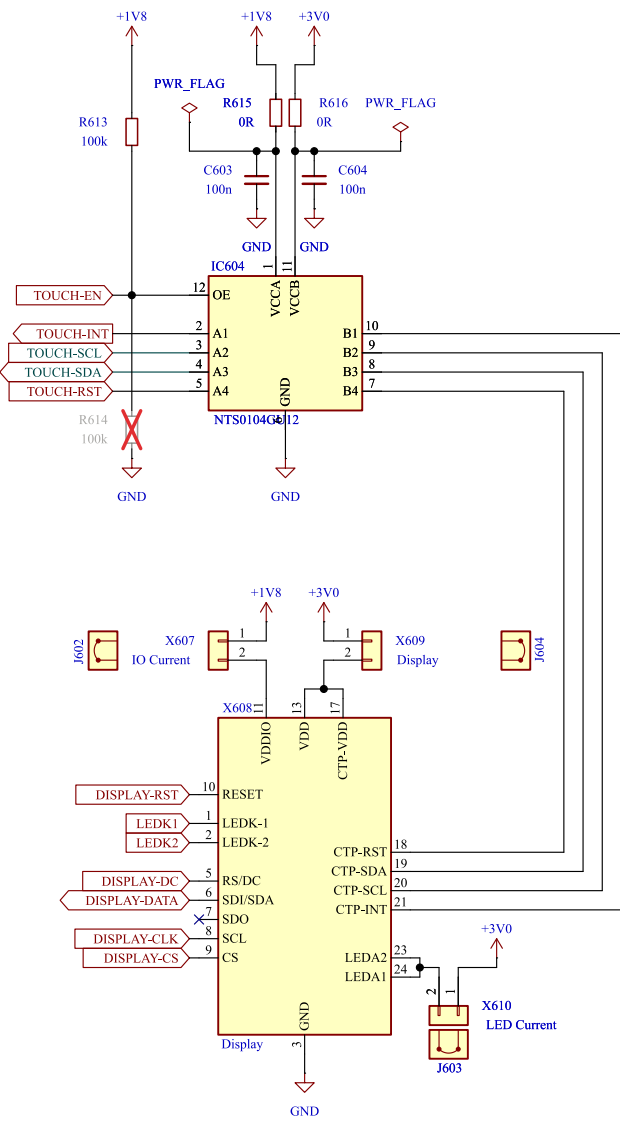


Comments:	Company: ZSWatch		Variant: PRELIMINARY	Git Hash: dbc49fa
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: Peripherals	File Name: Peripherals.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.2.2+ (Unreleased)
Sheet Path: /Project Architecture/Peripherals/		Reviewer:	Size: A3	Sheet: 5 of 7

Microcontroller
(NORA-B106/nRF5340)

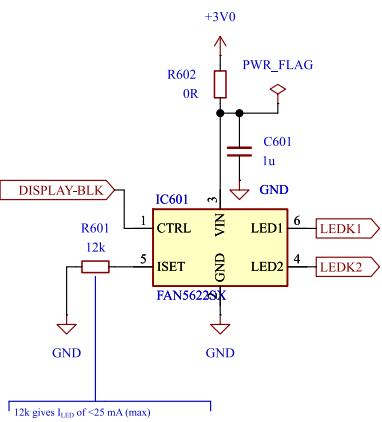


Display

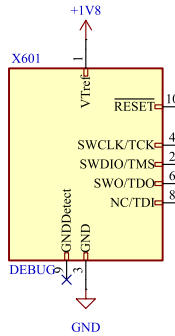


NOTE: 1V8 <=> 3V0 level shifter
NTS0104G012 has internal pullups. Works for I2C.
<https://www.nxp.com/docs/en/data-sheet/NTS0104.pdf>

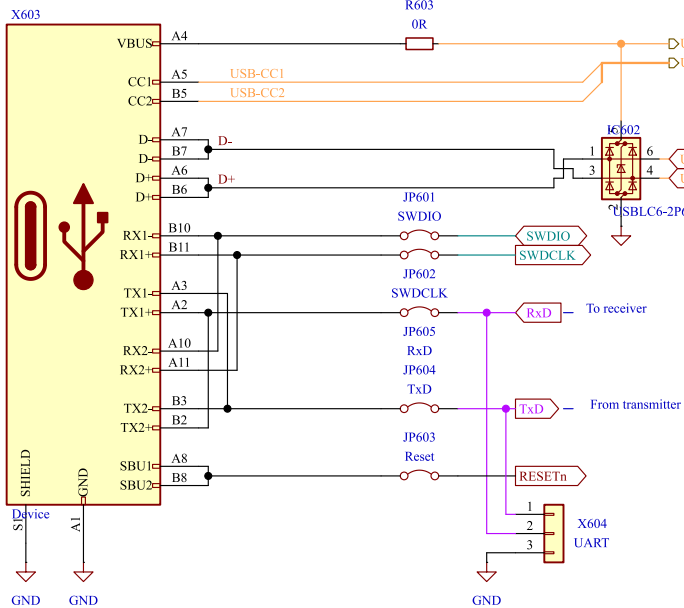
Display backlight



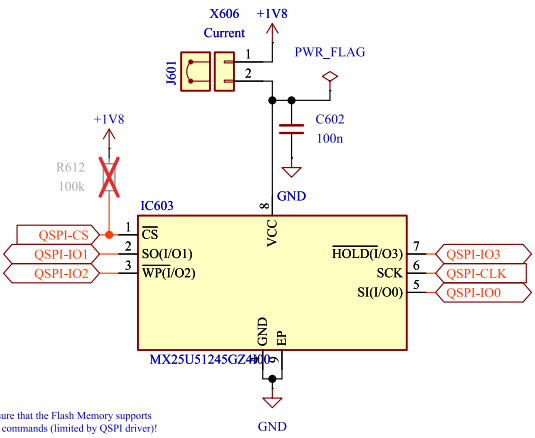
Debug connector



USB



External Flash



NOTE: Make sure that the Flash Memory supports
4k sector erase commands (limited by QSPI driver)!

Comments:	Company: ZSWatch		Variant: PRELIMINARY	Git Hash: dbc49fa
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: A3	Sheet: 6 of 7

[7] Revision-History

Version 1.2.0 - 2025-09-07

- Fixed
- Fix wrong orientation for SWD header (#117)
- Added
- Add pull-up resistor for display enable signal. Can be disabled via GPIO. (#123)
 - Add new KiBot template (#116)
 - Add orientation marker for IMU (#124)
 - Add power LEDs (#128)
 - Add reset button (#131)
 - Add testpoints for microphone I2S (#145)
- Changed
- Shrink down the PCB to save costs (#115)
 - Replace pin headers with THT to make them more robust (#121)
 - Rename "Power" jumpers to "Current" (#129)
 - Replace level shifter for RTC with PCA9306GF (#133)
 - Switch back to old LED driver (#134)
 - Power microphone from PMIC (#145)
- Removed
- Remove APDS-9306 interrupt (#122)

Version 1.2.1 - 2025-09-26

- Added
- Add missing pull-ups for I2C lines (#158)
- Fixed
- Fix wrong ship mode button (#155)
- Changed
- Change part number for SWD connector to 62201021121 (#157)
 - Set pull-up for Flash CS to NA (#158)
 - Center the mounting holes for the display (#159)

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	Comments:	Company: ZSWatch		Variant: PRELIMINARY	Git Hash: dbc49fa
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Revision-History	File Name: Revision History.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.2.2+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 7 of 7